

CSMFAB000000000106 V2.0

Strategic Architecture for High-Pressure Gas Infrastructure: Mitigating Telluric and Geomagnetic Threats

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- ASME B31.12-2025: hydrogen service updates for non-metallic pipe included
- ASTM D2996-2024: GFRP pipe approved to 1,500 psi for natural gas service
- Enbridge 2025 monitoring program: 147 CP anomalies documented during elevated Space Weather Index (SWI > 5) periods
- New corrosion rate equation: 10 kg steel dissolved per ampere-year of GIC discharge
- Electroosmotic flow control for improved soil-pipe interface management

1. The Steel Pipeline as Planetary Antenna

A natural gas transmission pipeline spanning 500 miles is a wire antenna 800 km long. In a 10 V/km geoelectric field (G3 storm):

$$I_{GIC} = \frac{E_{geo} \times L}{R_{pipe}} = \frac{10 \text{ V/km} \times 800 \text{ km}}{R_{loop}}$$

For 36-inch diameter steel pipe (wall 0.5 in, grade X70): $R_{loop} \approx 0.4 \Omega$ total

$$I_{GIC} = \frac{8,000}{0.4} = 20,000 \text{ A}$$

At 10 kg steel dissolved per ampere-year of anodic discharge:

$$m_{dissolved} = 10 \times I \times t = 10 \times 20,000 \times (1/8760) = \mathbf{22.8 \text{ kg/hour}}$$

During a 6-hour G3 storm: **137 kg of steel dissolved** from the pipeline network — equivalent to a 2-meter section of wall thickness perforating.

2. GFRP Transmission Pipe — V2.0 Approval Data

ASTM D2996-2024 (thermosetting filament-wound pipe for natural gas): - Maximum operating pressure: 1,500 psi (natural gas, updated 2024) - Aegis design target: 2,000 psi — requires 1.33 safety factor $\Rightarrow$ wall thickness increase of 18% - Hoop stress (Barlow equation): $\sigma_H = PD/2t$ - At 2,000 psi, D = 36 in: $t_{required} = 2,000 \times 36 / (2 \times 50,000) = 0.72$ inches $\Rightarrow$ achievable with GFRP

GFRP pipe properties (glass/epoxy filament wound): - Hoop tensile strength: 50,000 psi (vs 70,000 psi X70 steel — slightly lower, compensated by wall thickness) - Zero electrical conductivity $\Rightarrow$ zero GIC coupling, zero cathodic protection required - Zero corrosion $\Rightarrow$ no external dielectric coating needed $\Rightarrow$ \$4.50/m saved vs steel pipe protection

3. Alumina Ceramic Joint Fittings

At each joint (every 12 m), the GFRP pipe connects via alumina ceramic compression fittings:
- Flexural strength: 380 MPa (Al_2O_3 , 99.5% purity) - Rated for: 2,500 psi burst pressure (25% above design max) - Zero conductivity: joint-to-joint GIC path break confirmed at every 12 m segment - Sealing: PTFE-encapsulated O-ring, rated -40°C to $+200^\circ\text{C}$

4. Polymer Smart Pig (GPS/GIC-Immune Inspection)

Traditional steel smart pigs become magnetized/unreliable during GIC events. Aegis smart pig: - Shell: PEEK CF40 composite (magnetic-transparent, radiation-hard) - Inspection: Phased Array Ultrasound (PAUT) transducers in ceramic housing - Sensing: FBG optical strain gauges (no metallic element) - Power: on-board solid-state battery (LiFePO_4 , radiation-tolerant) - Data: stored in MRAM (magnetically immune) for post-inspection download